Chapter 9

Water pollution biology

1 Introduction

As discussed in Chapter 4, matter and energy move through trophic states within water systems and affect how the nonliving components of a system interact with the living components. There are countless interrelationships between abiotic and biotic systems, respectively. Thus, biogeochemical interactions can be studied, documented, and analyzed to track the earth's mass and energy balances, both thermodynamically and fluid dynamically [1]. Even the terrain of the ecosystem results in part due to biology. Physical and chemical processes make the original rock material suitable for microbial and root growth. Lichen and other simple plant systems further degrade solid rock and its unconsolidated materials, allowing roots of plants to take hold and expand.

Biota consist of fauna, flora, or microbial individual organisms and populations that can be receptors of water pollutant's effects via exposures. Pollutants are causative agents of environmental harm and human, animal, or plant diseases. Pollutants are stressors that can be transferred to, from, and within environmental media, e.g., a plant's roots extract a pollutant from the soil to plant tissues followed by the accumulation and transfer up to other trophic states from producers on to consumers in the food chain. Biota of various genera transform a pollutant biochemically within the organism, i.e., biotransformation, which can make a pollutant less toxic by depuration or more toxic by bioactivation. Biota can be seen as their own environmental compartment or part of the composition of other compartments, such as bacteria in river sediment and fish in the water column of an estuary.

As such, an organism can be envisioned as a control volume in which energy and matter enters and exists and where complex physical and chemical processes and mechanisms happen. Within the control volume, both abiotic chemical reactions and biochemical reactions take place continuously. Metabolic reactions are regularly catalyzed enzymatically [2,3] and mediated by microorganisms living within the organism [4–9]. Even the simplest one-cell organism is a chemical reactor in which molecules enter, are metabolized to produce energy and cellular molecules, and release new molecules as wastes from these processes [10–14].

The principal environmental hazard is toxicity, whether ecological toxicity[a] or human toxicity. The collective term, environmental toxicology, is inclusive of toxic substances in water, air, soil, sediment, food, or materials. The toxicity is differentiated by onset of an adverse event, e.g., cancer in humans or loss of species in a river from a fish kill. If the toxic response occurs soon after the exposure, it is an acute effect. If the adverse event occurs after a substantial amount of time, such as a cancer latency period, it is a chronic effect. Between the two, especially in health risk assessments, is a subchronic event.

Characterizing a hazard includes the sources, environmental conditions, and the time and dose needed to elicit an adverse event. These and other elements of risk are discussed in Chapter 17.

This chapter specifically addresses how organisms and biological systems affect water quality. Biological processes and mechanisms occur within organisms and in the organism's environment. From an engineering standpoint, these same biochemical can be used to remediate pollution [1,15–22].

a. Often shortened to "ecotoxicity."

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2 Biological organization

The water molecule configuration depicted in Fig. 4.1 in Chapter 4 is the basis for water pollution biology [23]. The cycle of biological processes includes the decomposition microbes that are part of the soil to plant life to the animals that consume plants to omnivores and carnivores, until they die and are decomposed by microbes. The cycle continues.

Thermodynamically, to overcome entropy, external energy must be added to keep biological processes proceeding. Indeed, life is constantly in need of new energy as it fights chaos [24]. Whereas physicists use thermodynamics to explain how mass and energy are distributed, ecologists consider the distributions and exchanges between mass and energy to explain the complex interrelationships between and within the compartments of the food webs and chains [24–26]. Humans are among the consumers [27]. Food chains represent the dynamics, complexity, and vulnerability of aquatic and other environmental systems (see Fig. 9.1). Species at a higher tropic level are predators of lower-trophic state species, which means that biotic matter and energy flow downward. The rates and quantity of the transfer of biomass and energy upwardly and downwardly to and from adjoining tropic levels vary in time and space. Stress from a pollutant entering an ecosystem alters these trophic interrelationships. For example, diminishing the overabundance of a primary producer like algae will modify predator-prey relationships, since the algal blooms may extract dissolved oxygen that in turn reduces the abundance of consumers and ultimately the top aquatic predators like seals and sharks, as well as top terrestrial predators like piscivorous birds, e.g., eagles and carnivorous mammals, including humans.

All organisms are made up of various molecular arrangements of the elements carbon (C), oxygen (O), hydrogen (H), and often nitrogen (N), in addition to various micronutrients. The C, O, H, and N are the biophile elements with an affinity for each other, which leads to their formation of complex organic compounds. Viruses, bacteria, and fungi are very efficient in using available organic material as sources of energy and carbon. Although a minority of microbial genera, pathogens have received most of the interest, at least from the general public, as agents of disease and, notably, epidemics and pandemics. However, the microbe is usually only one agent in the causal chains of diseases. Often, the cause of the epidemics involves numerous interrelationships among the trophic states. For example, when habitats are changed to allow advantages

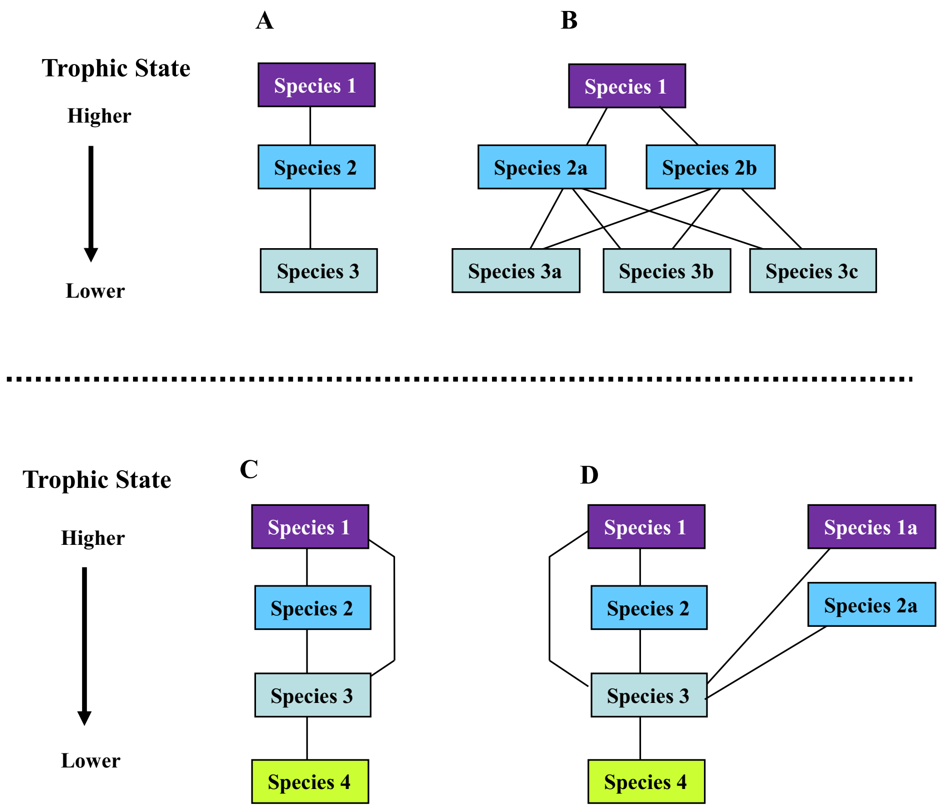

FIG. 9.1 Energy and matter flow in environmental systems from higher trophic levels to lower trophic levels. Lines represent interrelationships among species. (A) Linear biosystem; (B) Multilevel trophic biosystem; (C) Omnivorous biosystem; and (D) Multilevel biosystem with predation and omnivorous behaviors. An interference at any of these levels or changes in mass and energy flow can lead to disasters. *(From D. Vallero, Air Pollution Calculations: Quantifying Pollutant Formation, Transport, Transformation, Fate and Risks, Elsevier, 2019; used with permission.)*

to certain species, such as rats carrying the plague bacterium, *Yersinia pestis*, the epidemic cannot simply be attributed exclusively to the pathogen, but to the overall changes to the environment to allow the pathogen population to grow and to persist. Climatic shifts, poor water and air quality, and societal stressors like poverty, crime, and poor housing conditions can lead to outbreaks and worsen the outcomes of diseases in vulnerable human subpopulations [28–30].

Ecosystems and individual organisms and their components are thermodynamic systems. Beyond thermodynamics, a "system" is a method of organization, from smaller to larger aggregations. The "ecosystem" and the "organism" are examples of both connotations. They consist of physical phases and order (e.g., producer-consumer-decomposer; predator-prey; individual-association-community; or cell-tissue-organ-system). They also help to explain how matter and energy move and change within a parcel of matter. Within the context of thermodynamics, a system is a sector or region in space or some parcel of a sector that has at least one substance that is ordered into phases. Reactors, ground water plumes, the entire hydrosphere, communities, organisms, and cells have qualities of both closed and open systems. A closed system allows no material to enter nor to leave the system. The open system allows material to enter and to leave the systems, known as a control volume or, for engineers, a control mass [31].

Pollution results from complex interactions of substances in various states of matter in the various environmental compartments where substances are formed, transformed, and transported. The transformations may render a compound that was previously not inherently harmful to become toxic to one or more species in the ecosystem. Conversely, what may have been a precursor to a pollutant may have been altered to become a less harmful compound or compounds. During transport, a pollutant may also undergo chemical change in various parts of an ecosystem and have a selective effect on a habitat. These changes can be incremental and may not become obvious for years or decades.

3 Cells

Living things contain water, compounds of carbon, and deoxyribonucleic acid (DNA). These are all involved in cellular reproduction. Binary fission is the most common cellular reproductive mechanism (see Fig. 9.2), which begins with cell growth that causes the cell to enlarge and elongate to hold an increase in cellular components. The middle of the enlarged cell continues to enlarge and forms two progeny cells, each with a complete copy of the parent cell's genome and half of the cytoplasm. Cytoplasm is a gelatinous cellular liquid composed of water, salts, and organic molecules. Intracellular organelles, especially the nucleus and mitochondria, are bounded from the cytoplasm by membranes. Cytokinesis is the cytoplasmic division at the end of the cellular separation process. The replication of DNA begins where the chromosome is attached to the inner cell membrane. Ultimately, the cells separate completely [32].

Single-cell organisms are prokaryotes and taxonomically fall under the Bacteria and Archaea domains. The more complex eukaryotes fall under the Eukarya domain. All three domains contain unicellular organisms. Eukarya also contains multicellular organisms. Eukarya's single-cell organisms include fungi and protists. Protists include algae and protozoa like amoebas [32].

As mentioned, all cells contain DNA, and have cellular or plasma membranes, cytoplasm, and ribosomes, the organelles that manufacture proteins. A eukaryotic cell has a nucleus, whereas a prokaryotic cell does not. The prokaryotic cell's genetic material is suspended in the cell's cytoplasm.

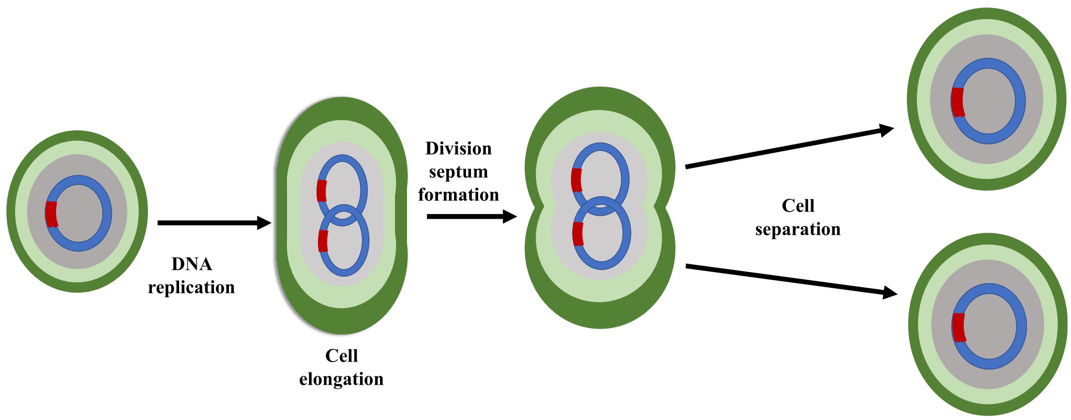

FIG. 9.2 Steps of cellular reproduction by binary fission. *(Data from D.A. Vallero, Biotic physics, in: Methods and Calculations in Environmental Physics, AIP Publishing LLC, Melville, New York, 2022, pp. 9-1–9-40.)*

Mitosis and meiosis are the two types of binary fission. Mitosis produces cells is the process shown in Fig. 9.2, in which a cell simply divides into two cells. Meiosis produces gametes that are needed to produce a cell or nucleus that contains two complete sets of chromosomes, i.e., a diploid. Chromosomes are threadlike structures comprised of nucleic acids and protein that carry genetic information to form genes made of DNA. Gametes have half of the genetic material needed to initiate a new diploid by fusion with a gamete from the opposite sex (see Fig. 9.3). The female gamete is an egg cell. The male gamete is a sperm. Most animals are diploid organisms, so that their somatic cells are diploid, and their gamete cells are produced through meiosis. Certain genera of insects are exceptions. In bees, wasps, and ants, the male is haploid because it develops from unfertilized eggs. Sexual reproduction distinguishes animals from other organisms like fungi, protists, and bacteria, that commonly or exclusively reproduce asexually. Some animals can also reproduce asexually, such as cnidarians, flatworm, and roundworms.

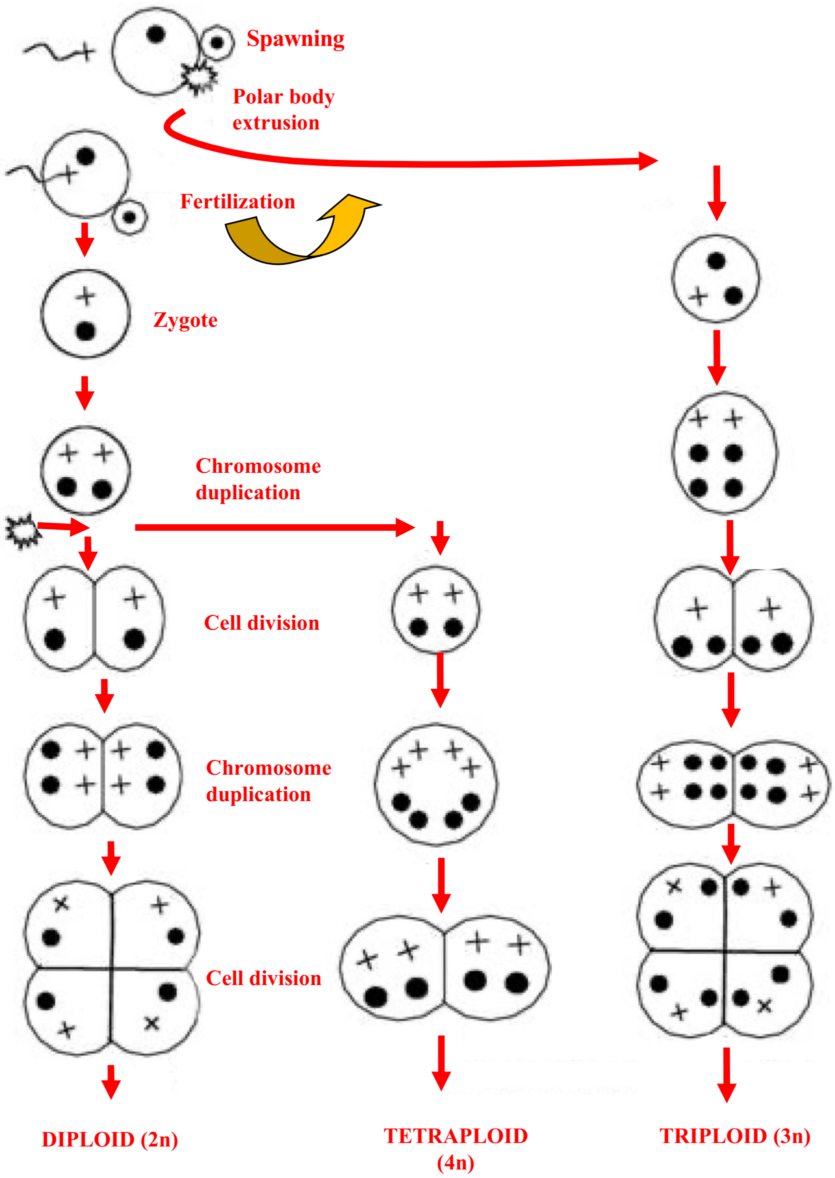

FIG. 9.3 Steps in gamete fertilization and cell division that lead to the normal diploid fish embryo. Induction of triploidy or tetraploidy occurs by for shocks (thermal, chemical or pressure) at a specific time after fertilization. Notes: denotes the point at which the shock is applied; ● denotes one haploid chromosome set derived from the female parent; + ● denotes one haploid chromosome set derived from the male. *(Data from National Research Council, Biological Confinement of Genetically Engineered Organisms, National Academies Press, 2004.)*

4 Invasive species

Invasion is special type of water pollution when opportunistic organisms can colonize a habitat. During the invasion, ecological structure and function are changed and become unbalanced, leading to the loss of diversity and other ecological damage. Invasive organisms can be any type of biota, that is, microbes, plants, and animals. Usually, their presence is the result of human activity.

An invasive organism has few or no predators during their entry into an ecosystem. As such, they can consume the food sources much faster than their native competitors [33]. For example, the Great Lakes basin has experienced major problems with invasive species. Many invasive species have been identified in the Laurentian Great Lakes drainage basin. Lake Erie watershed alone has 132 nonnative species, including 20 species of algae, 8 submerged plants species, 39 marsh plants species, 5 tree and shrub species, 3 pathogenic bacteria species, 12 mollusk species, 9 oligochaete species, 9 crustacean species, and 23 fish species. The increase is due in the most part to switching from solid to water ballast in cargo ships and the opening of the St. Lawrence Seaway in 1959. Among the nonnative invasive fish species recently invading Lake Erie is the Chinese bighead carp, *Hypophthalmichthys nobilis* [34].

One of the means of addressing invasive species is by using biocontrols, i.e., using one living organism to limit and eliminate another organism's effects. Introducing sterility has been used to control fish populations by interrupting the normal fish development process. The traditional sterilization approach change's the invasive fish's ploidy (i.e., affecting the complete set of chromosomes), by modifying the chromosome development using pressure, thermal, or chemical shock at the point of fertilization to disrupt the egg's normal extrusion of a polar body containing a haploid set of maternal chromosomes.

Inducing triploidy has been used effectively to produce sterile offspring by disrupting sexual reproduction (see Fig. 9.3). The retained polar body produces an embryo with two haploid chromosome sets from the female (rather than the normal one haploid set) as well as a third set from the male [35]. The odd sets of chromosomes appear to interfere with the mechanics of pairing of homologous chromosomes during each cell division, which disrupts normal gamete development. The resulting triploid varies from the normal diploid number of chromosomes. However, the major weakness of this method of sterilization is it can be incomplete, that is some of the offspring are not triploid. This can be ameliorated somewhat by producing tetraploids in the first generation.

An important environmental risk of this type of sterilization is the escape of these genetically modified organisms into the wild populations, since triploids still have sufficient levels of sex hormones to elicit normal courtship and spawning. Thus, the entire wild relative species' reproductive success would be in jeopardy, which is exacerbated if the wild relatives are endangered or threatened. In addition, given the large numbers of fish that escape fish farms, if many triploid transgenic fish enter the environment recurrently, some could survive to adversely affect fish diversity [35].

New genetic engineering methods have been developed to induce fish sterility, particularly the application of interference ribonucleic acid (RNAi), to produce what are known as sterile ferals [36]. The RNAi method involves the insertion of a transgene that blocks the expression of an endogenous gene needed to develop gametes and embryos. The blocker expression is controlled by an inducible promoter. This indicates that the transgenic approach has all the ecological concerns of traditional techniques, but with additional uncertainties, such as the presence of repressor chemicals in the water [36].

A biotechnological concern is the possible spread of deleterious genetic material. Controlling nonnative fish species invasions with transgenic fish of the same species as the targeted species is designed to be disruptive to the fish life cycle (see Fig. 9.3). These disruptions can occur from before the embryonic stage up to juvenile development. Several endogenous genes' expression can be disrupted in fish, including the aromatase, estrogen receptor, gonadotropin releasing hormone, protamine, vitellogenin and growth regulating genes.

Species can hybridize with closely related species, which increases the potential of transgene spread to near relatives of nontarget species. The amount of hazard and risk is a function of the proximity and availability of these near relatives, their proficiency to hybridize, their fertility, and viability of their offspring.

With transgenic species there is also the potential for unexpected expressions such as toxicity or allergenicity differences between the native and transgenic species. This could lead to human health effects. The bottom line is that the decision to use these technologies for even a worthwhile environmental effort (dealing with invaders), must be based on a thorough identification of these hazards and a reliable assessment of the risks that these technologies may elicit.

5 Plant physiology and pollution

Plants are very important to water quality, as they take in and release water and gases by their roots and leaves. In the water and air, various nutrients are dissolved and suspended, but so are other substances. In the leaves, pollutants can enter the plant via the stomata, which will interrupt processes in the chloroplasts. Chloroplasts are plant cell organelles that convert light energy into chemical energy via the photosynthesis. As such, they are needed to combine water and CO_2 in the presence of sunlight to form sugars and to release O_2:

$$6CO_2 + 6H_2O \xrightarrow[light]{} C_6H_{12}O_6 + 6O_2 \tag{9.1}$$

The sugar in this case is glucose ($C_6H_{12}O_6$). Transpiration moves water from roots to the leaves resulting in the emission of water vapor to the atmosphere. Nutrients move advectively and diffusively throughout the plant. Plants also respire to oxidize carbohydrates by O_2 to form CO_2 and H_2O and the release of energy, represented by adenosine triphosphate (ATP):

$$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + ATP \tag{9.2}$$

ATP is then hydrolyzed to adenosine diphosphate (ADP):

$$ATP + H_2O \rightarrow ADP + P_i + \text{free energy} \tag{9.3}$$

ADP then reacts with the phosphate molecule (P_i) to form ATP

$$ADP + P_i + \text{free energy} \rightarrow ATP + H_2O \tag{9.4}$$

Transpiration, photosynthesis, and respiration occur simultaneously as they transport O_2, CO_2, and H_2O throughout the plant. These mechanisms also move these and other chemical compounds into the troposphere, i.e., phytopartitioning. The presence of certain chemical compounds in plant tissues can cause direct or indirect harm to the plant, as well as adversely affect soil microbial populations, such as replacing beneficial bacteria with those that are toxic to the plant. Given that microbes are key decomposers that mineralize organic compounds needed by plants to survive and that first order consumers depend on specific plants, changes in microbial populations upset the balance of the entire ecosystem. Microorganisms already stressed from direct pollutant pollution, e.g., sulfur dioxide (SO_2), can be further stressed by ambient changes, such as droughts and fires. Foliar and root damage diminishes plant vitality, increasing the plant community's vulnerability to insect infestation, and weakens competition with weed species [23].

Approximately one-third of the earth's land surface has 20% or more crown tree cover [37]. Crown cover is measured as the canopy cover plus the amount of skylight obstructed by tree canopies. Human activities are particularly stressful to tropical rainforests. These include clearcutting, harvesting whole trees, and building roads that traverse habitats. Deforestation has global implications, e.g., concomitant increases the emissions of CO_2 and other greenhouse gases to the atmosphere. Soil and plants in tropical forests sequester 460–575 billion metric tons (tonnes) of carbon worldwide with each square kilometer of tropical forest holding 45,000 tonnes of carbon. From 1850 to 1990, deforestation released 122 billion tonnes of carbon into the atmosphere. Currently the deforestation carbon emission rate is 1.6 billion tonnes per year [38]. A secondary air pollution impact of deforestation is the increased emissions of fugitive dust [39].

Water uptake and storage is the key feature of the physics and biochemistry of vascular plants. As a solvent and medium for reactants and suspended matter, water provides the site for plant metabolism to builds cells. Water is the principal vehicle for heat dissipation within the plant and provides temperature homeostasis. Excess water vapor decreases the density of air, promoting circulation and assists in the availability of CO_2 for photosynthesis, in which water also acts as a reactant and reagent. Starches, for example, react directly with water, i.e., under hydrolysis, during metabolism resulting in sugars needed for healthy plant physiology. Water also serves physical needs, e.g., sufficient water content is needed to maintain turgidity, i.e., cell wall expansion and the osmoregulatory function of the vacuole to prevent wilting [40]. The plant must optimize available moisture as root uptake of water increases a vascular plant's water content, as transpiration simultaneously decreases water content [41].

Upon entry to the root, dissolved and suspended nutrients and other substances flow into the plant and enter several layers of differentiated cells. The water crosses these layers successively, i.e., composite transport [42], via pathways composed of cell walls and intercellular space (apoplastic pathway), plasmota (symplastic pathway), or water channels (osmotic, transcellular pathway) [43].

In addition to molecules of nutrients and pollutants, plants also take up ions. Ionic transport occurs between soil water and roots by active pumping across the root cell cytoplasm's plasma membrane. Conversely, during transpiration, roots allow water to pass through them, rather than being taken up like ions. The water potential (ϕ_T) gradient makes this movement possible [43–45]. The total water potential (ϕ_T) combines several potentials [46]:

$$\phi_T = \phi_p + \phi_m + \phi_z + \phi_o + \phi_{ob} + \ldots\phi_n \tag{9.5}$$

Where, ϕ_p is the pressure potential, ϕ_m is the matric potential, ϕ_z is the gravitational potential, ϕ_o is the osmotic potential, ϕ_{ob} is the overburden potential, and ϕ_n represents the potentials induced by other factors.

Pressure potential is the height of water above a designated point, i.e., the water pressure exerted by the saturated column of water above that point. Matric potential owes to capillary and sorptive forces. Gravitational potential is caused by the force of gravity measured from a reference point. Osmotic pressure is the solute pressure caused by water's attraction to salts dissolved in the water film around particles. Overburden potential is caused by the weight of soil on a given point in the soil column holding the roots [46,47].

Composite transport is affected by different types of movement through individual cells and tissues, with preferable routes for flow of low resistance controlled by the plant's geometry, properties of cell walls, and permeability of the various layers [43–45].

When entering the xylem, the transport becomes more extensive than in the root, with vessels carrying the water via the longest reaches of the plant. In a corn plant, for example, xylem hydraulics cause how water to be transported several meters compared to only a few millimeters distance from the root surface to the xylem. Likewise, the distance from the xylem to mesophyll cells is but a few millimeters. Thus, xylem transport is about three orders of magnitude longer [43]. The major limitations of xylem flow are friction and cavitation, according to Poiseuille's law and probability theory, respectively. Poiseuille's law states that the velocity of a fluid's steady flow via a narrow tube like the xylem varies directly as the pressure and the fourth power of the radius of the tube (r), and inversely as the length of the tube (L) and the coefficient of viscosity. Thus, the pressure drop is directly proportional to the length of the xylem vessel and volumetric flow rate, and inversely proportional to the xylem vessel radius and cross-sectional area [48,49]:

$$\triangle p = \frac{8\mu LQ}{\pi r^4} = \frac{8\pi\mu LQ}{A^2} \tag{9.6}$$

Where, $\triangle p$ is the pressure difference between the two ends of the tube, μ is the dynamic viscosity, Q is the volumetric flow rate, and A is the cross-sectional area of the tube.

The formation of vapor-filled cavities, bubbles, and voids in a liquid, known as cavitation, is caused by static pressure that reduces the liquid's vapor pressure (see Discussion Box 9.1). Xylem cavitation results when the water tension inside the xylem increase enough that dissolved air within water expands to fill either an entire vessel or the xylem's tracheids, i.e., water-conducting cells that have no perforations in their cell walls. Embolism is the blocking of a xylem vessel or tracheid by an air bubble or cavity [60]. Subsequently being sorbed by the root, water reaches and crosses the epidermis. This flow is in the direction of the center of the root, traversing the cortex and endodermis, until it reaches the xylem [61].

5.1 Diffusive flux

The time that substance stays within plant tissue is the plant's mean residence time (MRT or τ):

$$\tau = \frac{M}{J_{out}} \tag{9.7}$$

Where, τ = MRT at steady state, M is the mass of the chemical species in a plant, and J_{out} is the mass of the chemical species leaving the plant. The concentration gradient (i_c) against the resistance of the chemical compound to exit the plant is described by Fick's law of diffusion:

$$J = \frac{i_c}{R} \tag{9.8}$$

Where, R is the resistance of the plant tissue. The exchange of gases between a plant's leaf and the atmosphere is chiefly determined by the size of the opening of stomata (Fig. 9.4), i.e., small pores in the plant's epidermis that are the sites of vapor and gas flux, mainly water vapor and carbon dioxide (CO_2). The stomata are "gatekeepers" for gas diffusions between the ambient air and the plant that balance gas exchange between the leaf interior and the bulk atmosphere [62]. The stomata limit transpiration by maximizing CO_2 flux into the leaf and minimizing H_2O flux out of the leaf. The gas flux is governed by a pathway consisting of a series of R values, beginning with the interface between the leaf and the air, i.e., the leaf boundary layer. The stomatal pore openings make available the dominant R between the air and the interior of the leaf. Simultaneously, CO_2 provides additional R as it encounters the aqueous and lipid boundaries within the leaf tissue, i.e., mesophyll cell and chloroplasts. Collectively, these processes comprise the mesophyll resistance. The outward flux of H_2O to the air results from all the forces in the R series except for mesophyll resistance. Guard cells regulate stomatal pores that expand or contract in response to external stimuli, like changes in temperature and humidity.

Discussion Box 9.1 Vapor pressure and pollution

From a water pollution perspective, arguably the most useful expression of vapor pressure is that it is the pressure at which a liquid and its vapor are in equilibrium at a given temperature. The vapor is pushing against the atmosphere, so it is one way a compound exits a waterbody. The greater the vapor pressure the faster liquids evaporate. A liquid is at its boiling point when vapor pressure reaches atmospheric pressure. Vapor pressure is measured in pressure units, including atmospheres (atm), millimeters of mercury (mmHg) or kilopascals (kPa). As a point of reference, normal atmospheric pressure is 1 atm (760 mmHg or 101.325 kPa).

A substance like molecular nitrogen (N_2) and oxygen (O_2) that is always in the gas phase at or near the earth's surface under typical conditions, i.e., permanent gases, has a vapor pressure of 1 atm (atm), which is 760 Torr or 101.325 kPa. Other compounds behave like permanent gases under environmental conditions, such as methane (CH_4), which becomes a liquid at about minus 82 °C, which is only reached occasionally in Antarctica. However, many compounds can be found in all three physical phases in the environment, so they are often classified according to their likelihood to reach the gas phase.

Vapor pressure is commonly used to categorize pollutants that are likely to move to the gas phase under environmental conditions. For example, volatile organic compounds (VOCs) readily move to the headspace of a container or to the atmosphere. Generally, these compounds have vapor pressures greater than 10^{-2} kPa. The semivolatile organic compounds (SVOCs) have vapor pressures between 10^{-5} and 10^{-2} kPa [50]. Others have proposed alternate ways to classify VOCs, such as evaporation rates over time, e.g., what remains after 6 months [51]. These values correspond to classifications that are based upon observations of the compounds' behaviors during air sampling [52].

Consider a scenario where 10 L of dichloromethane (CH_2Cl_2) is spilled to form 2 m^2 puddle of CH_2Cl_2. A slight 4 km h^{-1} wind is blowing. The temperature is 20 °C. Dichloromethane is a toxic organic solvent liquid with a vapor pressure of 0.46 atm at 25 °C.

Surface area is important an important factor because the amount of surface determines the rate of evasion due to vapor pressure. The greater amount of surface area, the greater the mass of the chemical compound that will exit the water. Temperature is also an important factor in vapor pressure since the greater the temperature, the more of the chemical compound that will leave the water.

Wind is also a factor because vapor pressure results from a kinetic process at the air-liquid interface that is driven by molecular interactions that allow energetic gas molecules in the air to transfer energy to volatile liquid molecules. This provides the liquid molecules of the chemical compound sufficient energy to change phase to a gas state. Increasing wind speed increases the probability of more energetic molecular interactions because a greater number of molecules traversing the surface of the standing water per time and therefore more chances for interaction.

Based on the scenario data, dichloromethane would vaporize relatively quickly, much faster than water will evaporate. Indeed, it would take about a half-hour for 10 L to completely evaporate under these conditions. Conversely, a 10 L-puddle of water would take hours to evaporate completely. The more volatile ethyl mercaptan (C_2H_6S; vapor pressure = 0.59 atm at 25 °C) would evaporate in about 17 min. The low vapor pressure ethylene glycol ($C_2H_6O_2$; 6 $\times 10^{-5}$ atm at 25 °C) takes quite long to vaporize. Thus, most of the spill will persist until it is physically removed. However, other processes, including microbial [52] and photochemical [53] degradation, could break down some of the compound while it is on the surface. Unfortunately, if not remediated, some will infiltrate the soil and can contaminate ground water. Any of these chemical compounds' evasion rate would be slower due to other factors, especially if the compounds sorb to soil particles and other surfaces, if temperatures decrease, and if they are dissolved by dissolved humic substances (DHS) and organic solvents that have lower vapor pressures [54].

Pollution control and handling systems for volatile pollutants must include design and operation criteria that are based on their propensity to transfer to and from the soil, water, and air. Compounds can be deposited onto terrestrial and water surfaces if they are dissolved in raindrops and other precipitation or sorbed to particulate matter. Conversely, volatile compounds can be released into surface and ground water and can subsequently become air pollutants as they migrate toward the atmosphere. The SVOCs especially comprise some of the most toxic, bioaccumulating, and persistent pollutants. A particularly important aspect of SVOCs is that they may be transported from soil in the gas phase or as aerosols [55]. Thus, if SVOCs are present in the waste stream such as in a municipal landfill or hazardous waste storage facility, they must monitored as both in the gas phase and in and on soil and particulate matter.

One substantial source of VOCs and SVOCs is their emission to the atmosphere during thermal treatment of solid and hazardous wastes. For example, halogenated dioxins are generated when precursor compounds like pentachlorophenol and polyvinyl chloride are in wastes that is incinerated at certain temperatures [56–58]. In addition, ash and residues contain products of incomplete combustion, such as polycyclic aromatic hydrocarbons, dioxins, furans, and hexachlorobenzene, as well as many pesticides, phenolics, solvents, and pharmaceuticals, which fall under the SVOC chemical classification. Proper management of the life cycle of these chemical compounds requires an understanding of the factors that lead to the factors involved in their release, movement, and degradation [31,41,59].

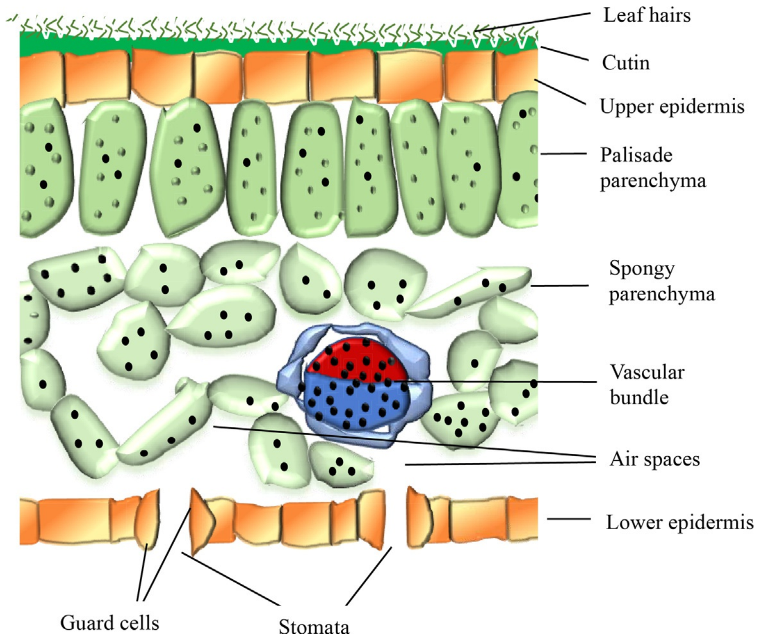

FIG. 9.4 Leaf cross-section. *(Data from D.A. Vallero, Fundamentals of Air Pollution, fifth ed., Elsevier Academic Press, Waltham, MA, 2014, p. 999 pages cm.)*

Stomatal pores make up only about 3% of the leaf surface, while being responsible for about 98% of CO_2 inward flux and H_2O outward flux [62].

Diffusive flux affects and is part of many biological processes, such as delivering carbon dioxide (CO_2) to a plant to begin photosynthesis.

Leaf flux example 1

Envision a plant leaf that contains CO_2 concentrations ranging from about 100 μL L^{-1} inside the leaf to 400 μL L^{-1} in the atmosphere [63]. The water vapor gradient of concentration ranges from saturation inside the leaf to the vapor pressure of the atmosphere. In this case, the vapor pressure is the point of equilibrium between the water in the plant and the atmosphere, i.e., when the water molecules leaving the leaf as a vapor equal the water molecules entering the plant as a liquid. The atmospheric vapor pressure is a function of the relative humidity and temperature of the air around the leaf. If the stomatal resistance is 5 s cm^{-1}, determine the flux of CO_2 into the leaf and water vapor exiting the leaf at 25 °C and 30% relative humidity.

First, calculate the amount CO_2 entering the leaf:

$$J = \frac{400 - 100\ \mu\text{L L}^{-1}}{5\text{scm}^{-1.}}$$

Convert the liters to cubic centimeters:

$$J = \frac{300\ \mu\text{L} \times 1000\ \text{cm}^{-3}}{5\text{scm}^{-1}} = 0.06\ \mu\text{L cm}^{-2}\ \text{s}^{-1}$$

Thus, by applying the gas laws, the amount of CO_2 entering the leaf is:

$$0.06\ \mu\text{L cm}^{-2}\ \text{s}^{-1} \times \frac{1\ \mu\text{mol}}{22.4\ \mu\text{mol}} = 0.0027\ \mu\text{mol CO}_2\ \text{cm}^{-2}\ \text{s}^{-1}.$$

Leaf flux example 2

Leaf flux is affected by many environmental conditions, including those of the water, soil, and air. For example, plants can be damaged in the summer from exposure to ozone (O_3), which forms when hydrocarbons and oxides of nitrogen react in the presence of sunlight. Thus, O_3 episodes usually occur during growing seasons. If the atmospheric concentration of O_3 increases and causes the stomatal aperture to shrink, which increases the resistance to 10 s cm^{-1}, what is the flux of CO_2 into the leaf and H_2O transpiration at 100% relative humidity and 25 °C?

Answer: The increased resistance means that the flux is halved, i.e., 0.0013 μmol CO_2 cm^{-2} s^{-1}. At 100% relative humidity and 25 °C, the air contains:

$$\frac{29.8\ \mathrm{g\ m^{-3}\ 1\ mol\ water}}{18\ \mathrm{g}} = 1.6\ \mathrm{mol\,m^{-3}}.$$

Thus, if the air has 100% relative humidity at this temperature, the water vapor content would be 33% of 1.6, i.e., about 0.5 mol m^{-3}, so the leaf-to-air flux is:

$$\frac{1.6 - 0.50\ \mathrm{mol\ m^{-3}} \times 1\ \mathrm{m^3}}{5\mathrm{scm^{-1}} \times 10^6\ \mathrm{cm^3}} = 0.23\ \mu\mathrm{mol\ H_2O\ cm^{-2}\ s^{-1}}.$$

The gas flux of any pollutant in vapor or gas phase follows the same pathways as CO_2 and water vapor. As shown in the example, elevated levels of tropospheric O_3 can damage crop yields by accelerating leaf senescence and detachment, i.e., abscission and by decreasing stomatal aperture size. These processes weaken carbon uptake and decrease photosynthetic carbon fixation. Elevated concentrations of O_3 depress yields of crops important to global food supply, e.g., average global yield reductions as high as 5.5% for corn, 14% for wheat and 15% for soybeans [64]. Thus, increasing emissions of hydrocarbons and oxides of nitrogen in the atmosphere will concomitantly increase atmospheric O_3 concentrations that induce stress on vascular plants, injure foliage, and decrease the sustainability of major food supplies. The stress can result in several problems, including [31]:

- Moderation of biomass growth by changing the amount of available carbon and by disrupting the translocation of fixed carbon to grains, fruits, pods, roots;
- Decrease in nutrient availability by redirection to synthesis of chemical protectants, or reduced transport capabilities by the phloem;
- Diminished carbon transport to roots that reduces nutrient and water uptake and affects anchorage;
- Harm to flowering;
- Pollen sterility;
- Ovule and/or grain abortion; and;
- Diminished ability of certain genotypes to withstand other environmental stresses like drought, high vapor pressure deficit, and increased photon ($h\nu$) flux density (i.e., larger numbers of photons passing through the stomata per second, $h\nu\ s^{-1}$) due to effects on stomatal control.

This is but another example of how pollution is a multicompartmental challenge. Water and soil pollutants can harm a plant at the root zone and within the plant. Malfunctioning stomata can result in substantial plant tissue loss. Atmospheric exchanges of sulfur dioxide (SO_2), oxides of nitrogen (NO_x) O_3, peroxyacetyl nitrate (PAN), and hydrocarbons with plant leaves are notable examples. Indeed, exposure to O_3 causes up to 90% of the air pollution injury to vegetation in the United States [65].

Generally, the pollutant gases follow the same diffusive pathways as CO_2, but are often followed by additional mechanisms, e.g., NO_x dissolves in cells and generates phytotoxic nitrite ions (NO_2) and nitrate ions ($NO_3{}^-$), which affect metabolism. Exposure to SO_2 closes stomata, which protects against further leaf injury but inhibits photosynthesis by diminishing CO_2 flux. Upon entering the plant, SO_2 dissolves in water and generates phytotoxic sulfite ions (SO_3^{2-}). Conversely, low concentrations of these ions can be beneficial by detoxifying the plant and providing sulfur, which is a plant nutrient. This is an example of an optimal range, below which is nutrient deficiency, and above which is toxicity. Of course, pollution-related crop loss varies in time and space, so assigning the fraction due to pollution is unclear. For example, when average daily ozone concentrations reach >50 ppb, yields of vegetables can be reduced by up to 15% [65]. Drought conditions that can worsen already threatened crops [23].

Root water uptake can be modeled. The Gardner model [66] assumes diffusivity is responsible for the uptake:

$$\frac{\partial \theta}{\partial t} = \frac{1}{r} \times \frac{\partial}{\partial r}\left[rD\frac{\partial \theta}{\partial t}\right] \tag{9.9}$$

Where, θ is the soil's volumetric water content, D is the soil diffusivity, t is time, and r is the radial distance from the root axis. The model has several simplifying assumptions, especially that every root is a cylinder of infinite length and of uniform radius, that all of the roots are separated by twice the radius of the cylinder, and that the water in the root moves only in the radial direction, not affected by gravity. The model also assumes that the initial soil water content (θ_0) is a uniform value that corresponds to the matric potential (Φ_0) in the soil. The initial conditions are [46]:

$$\theta = \theta_0; \Phi = \Phi_0; t = 0 \tag{9.10}$$

And, the boundary conditions are [46]:

$$2\pi\left(\frac{d\Phi}{dr}\right) = 2\pi D\left(\frac{d\theta}{dr}\right) = q, \quad r = a \tag{9.11}$$

Where, $\pi \approx 3.14$, a is the root radius, K is the capillary conductivity of the soil, and q is the water uptake rate by the root as the volume of water per unit length of root per unit time (units in $L^3/L/T$).

Other root water models account for some of the numerous relationships between roots and soils. For example, the Nimah and Hanks model accounts for the effect of salts on the soil and water potentials:

$$S(z,t) = \frac{[\Phi_{\text{root}} + (R_{\text{root}} \times z) - \Phi(z,t) - \Phi_n(z,t)]R(z) \times K(\theta)}{\Delta x \times \Delta x} \tag{9.12}$$

Where, Φ_{root} is the effective water potential in the root at the surface of the soil, soil depth (z) is zero, R_c is the flow coefficient in the root, which the model assumes to equal 0.05, and R_{root} is root resistance, which is equal to $1+R_c$.

Soil-plant-atmosphere-continuum (SPAC) systems assume downward water flows in a transpiration stream in a potential energy gradient form soil to root, from root to stem, and from stem to leaf, from which the water diffuses to the air [46,67]:

$$S_w = \frac{\Phi_{soil} + \Phi_{plant}}{R_{soil} + R_{roots}} \tag{9.13}$$

Where, S_w is the water extraction rate (flow) from a unit volume of soil, Φ_{plant} is the potential in the plant, R_{soil} and R_{roots} are resistances presented by the soil and roots, respectively, and Φ_{soil} is the total soil water potential, i.e., sum of matric (Φ_m), gravitational (Φ_g), and osmotic (Φ_π) potentials.

This movement is analogous to Ohm's law:

$$i = \frac{V}{R} \tag{9.14}$$

Where, R is the resistance, i is the flow of electrons, and V is the voltage. SPAC systems consider V to analogous to the water potential difference driving flow (S_w) and i is analogous to the flow of water across and within a plant segment [68]. Thus, the Ohm's law analogy allows the hydraulic conductance to be quantified as the inverse of hydraulic R of roots, stems, or leaves. It can also be used to quantify the hydraulic conductance from the soil to atmosphere through the whole plant.

These calculations provide an opportunity to introduce the concept of water-use efficiency (WUE), which is the ratio of water being used for plant metabolism to water lost to transpiration:

$$\text{WUE} = \frac{[C_{\text{CO2-fixed}}]}{[C_{\text{H2O-lost}}]} \tag{9.15}$$

Where $[C_{\text{CO2-fixed}}]$ is the mmoles of CO_2 fixed in the plant and $[C_{\text{H2O-lost}}]$ is moles of H_2O lost to evapotranspiration.

Elevated rates of evaporation lead to moles of water lost, which means they are not available for plant processes, especially photosynthesis. Thus, a declining WUE value indicates that stress is occurring or portends plant stress. For most vegetation, WUE ranges from 0.86 to 1.50, and varies by environmental conditions, with WUE increasing with decreasing stomatal conductance [69,70]. Stomatal conductance (g_s) is the rate of passage of CO_2 entering or water vapor exiting via the stomata of a leaf, usually measured in mmol m$^-$ s$^-$, a unit of diffusive flux [71].

Leaf flux example 3

In the previous example of an ozone air pollution episode, the leaf resistance increased to 10 s cm^{-1}. Calculate the flux of CO_2 into the leaf and H_2O transpiration at 100% relative humidity and 25 °C.

Recall that the stomatal resistance (r) was 5 s cm$^-$. Thus, the calculations show the water-use-efficiency (WUE) of this leaf when $r = 5$ s cm$^-$, is about $\frac{0.0100\ \mu\text{mol}}{\mu\text{mo}} = 10$ mmol mol^{-1}. The gradient of water vapor between the leaf and the atmosphere is much steeper than the gradient for CO_2, meaning that the WUE is very low, and plants are less efficient at holding fixed CO_2, than the atmosphere is at extracting the water from the plant by diffusion. Thus, as tropospheric CO_2 rises, the rate of photosynthesis declines.

6 Applied biotics

Biotic cells convert nutrients and energy into biomass through the processes of photosynthesis, metabolism, and ion exchanges in microbes and plants [1,73,73]. Consequently, each organism metabolizes chemical compounds for energy and to build cells, but for pollution degradation the microbial decomposers are the main organisms that break down larger molecules into simpler ones. Bacteria, protozoa, fungi, algae, and even viruses can degrade organic molecules. Organisms at the lower trophic levels transform chemical compounds in the direction of mineralization, from large molecules like amino acids, proteins, and lipids, to become simpler and ultimately inorganic compounds that lack carbon-to-carbon and carbon-to-hydrogen covalent bonds (see Fig. 4.14 in Chapter 4). In the opposite direction, at higher levels of biological organization, consumers transform smaller molecules into larger ones, with producers doing some of each (see Fig. 9.1) [74].

Microbes decompose the accumulation of biomass in an ecosystem. Wastewater treatment engineers mimic these processes in wastewater treatment and hazardous waste treatment facilities and other bioreactors (see Chapters 14 and 16) [1,10–12,19,20,77–77]. These mimicked processes include physical and mechanical attributes to enhance biotransformation, such as the passage of water and air through pores unconsolidated media in an engineered trickling filter system. The rates of microbial growth is determined by physical factors like Fick's law of diffusion, coefficients for sorption, solubility and precipitation of the solutes that combine with the surface tension and capillarity of the media to determine the ability of the microbes to adhere to the surfaces of the media [79,79]. Flow of the wastewater through the pores in the trickling filter is affected by temperature, vapor pressure of the water's constituents, atmospheric pressure, and thickness of the biofilm. Thus, applied biotics combines biological factors with physicochemical factors. Important physicochemical factors include the polarity of the chemical compounds that form the solutes in the wastewater mixtures, hydrogen ion concentrations in the wastewater, i.e., pH, temperature, and catalysts determine the microbial degradation rate laws, e.g., zero, first, second reactions [81,81].

An extracellular polymeric substance (EPS) mediate microbial attachment [83,83]. The EPS adheres as a slime layer to the cell in a capsule or onto surrounding so that the bacteria have an anchor from which to degrade the organic content of the wastewater, i.e., to feed [79,83]. Biofilms become denser with depth due to decreasing pore volume [79].

Engineers and scientists have observed that microbes' ability to break down organic chemicals as a source of food and metabolic processes can be applied at the facility-scale by acclimating the microbes and limiting their food to other compounds, namely pollutants. As evidence, the sanitary engineering pioneer, William Paul Gerhard (1854–1927), took note of the importance of microbial degradation for pollution treatment as early as the year 1909 [84]:

> *The purifying action, which takes place in the two stages of treatment mentioned, is due to the different kinds of bacteria, known respectively as the aerobic, the anaerobic and the facultative bacteria. None of the methods of treatment are automatic, and all require intelligent management and constant supervision.*

Microbial metabolic processes of oxidation, reduction, hydrolysis, and other mechanisms could be tailored to degrade increasingly complex chemical compounds [10–12,85]. Microbial population can adapt to conditions of food scarcity and stress. Soil bacteria like *Pseudomonas* spp. acclimate to exposure to pollutants by withholding their preferred food sources, so that the microbes must adapt metabolic processes to use the pollutants instead for energy [86].[b]

Pollutants can be transformed biologically by three specific mechanisms: use of the compound as an electron acceptor; use of the compound as an electron donor; or co-metabolism. The processes can occur together and simultaneously in the same system, depending mainly on the availability of molecular oxygen in various regions of the system [1].

Electron acceptance occurs as redox conditions approach reduction. Reductive biotransformation requires a source of carbon as the electron donor for the growth and metabolism of an anaerobic bacterium. For example, if the compound contains chlorine (Cl), reductive dehalogenation causes the Cl to bind to carbon while other elements are removed. This is a common way to decrease the toxicity of a chlorinated compound like polychlorinated biphenyls (PCBs). When O_2 is depleted as the terminal electron acceptor (TEA), i.e., a compound that receives an electron during oxidation of a carbon source. Numerous anaerobic bacteria can use another TEA to generate intracellular energy [87]. Various types of organic matter can become sources of the electron donor carbon during cellular respiration or photosynthesis.

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b. I recall in a wastewater treatment course at the University of Kansas, Professor Ross McKinney told us that the solution to most water pollution problems is "right under our feet." He meant that literally. Soil bacteria have the potential to destroy most organic pollutants; they just need a nudge. It has been a blessing to the environmental engineering community that Dr. McKinney decided to use his doctorate in microbiology from the prestigious Massachusetts Institute of Technology (MIT) to become a pioneer in applied microbiology, which today is better known as environmental biotechnology.

Electron donation occurs when a compound is used as the primary substrate, i.e., the electron donor, to allow the microbe to gain energy and organic carbon from the compound. The electron donor is a reducing agent, i.e., when it donates electrons it is oxidizing itself. Donation occurs under either aerobic or under certain anaerobic conditions. Lesser oxidized compounds like vinyl chloride or dichloroethylene are commonly biotransformed by electron donation.

When a compound is degraded co-metabolically, an enzyme already available in the microbe for other processes fortuitously catalyzes the reaction. The microbe has no benefit from the degradation of the compound, which indeed could harm or inhibit growth and metabolism a microbe. Co-metabolism has been most often observed in aerobic environments, but has potential under anaerobic conditions, as well [2].

The three degradation processes all involve reduction-oxidation reactions, i.e., redox, which is a key factor in the type of bacteria present in natural and engineered systems. As molecular oxygen (O_2^-) decreases in the consolidated media profile, the increasingly hypoxic conditions become less favorable for aerobes and the population of anaerobic bacteria increase. Wastewater, like much of the earth's biomass contains sulfur (S), so the low O_2 redox conditions induce bacteria-mediated reduction of oxidized forms to hydrogen sulfide (H_2S), with sulfate (SO_2^-) as the electron acceptor:

$$6SO_2^- + 37.5H^+ + 30e^- \rightarrow 3.75H_2S + 15H_2O \tag{9.16}$$

As discussed in Chapter 6, microbial populations degrade substrates and grow according to the empirically derived Monod equation:

$$\mu = \frac{\mu_{\max} S}{K_s + S} \tag{9.17}$$

Where μ = the specific growth rate of the microbe; $\mu_{\max}$ equals the maximum specific growth rate; S = concentration of limiting substrate; and K_s is the Monod growth rate coefficient representing the substrate concentration at which the growth rate is half the maximum rate. Recall that the $\mu_{\max}$ is reached at the higher ranges of substrate concentrations. K_s is an expression of the microbe's affinity for a nutrient. As K_s decreases, the greater the affinity microbe will have for that nutrient, as expressed by the correspondingly increasing μ.

The Monod-constant K typically ranges between about 1 and 10 g m^{-3} for organic substrates, but is much lower for O_2, i.e., $K \approx 1$ g m^{-3}. Given identical environmental conditions, the K will follow a zero-order growth curve (see Fig. 9.5). K will change according to changes in environmental factors like nutrient concentrations, pH, temperature, and concentrations of substances toxic to the species of bacteria [17].

For a microbe to gain energy from its food, e.g., wastewater, the microbes need to have contact with it. Engineers enhance this degradation environment, such as by injecting activated sludge and adding activated microbes. As these conditions change, the microbes undergo the series of stages discussed in Chapter 6 [12]:

1. Lag phase.
2. Accelerated growth rate phase.

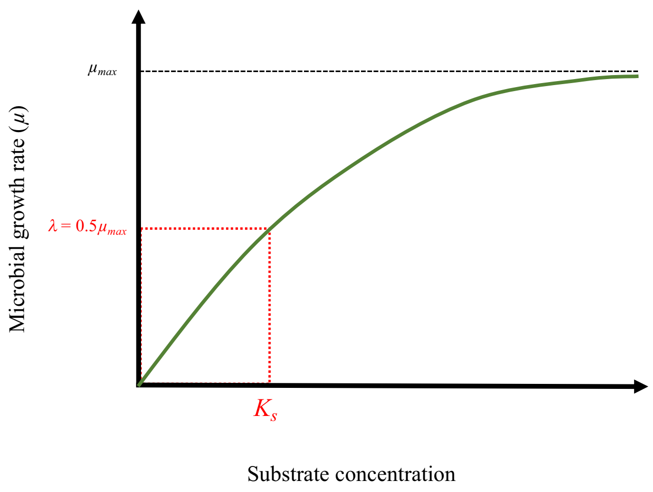

FIG. 9.5 Monod growth rate coefficient derived from many empirical studies showing the hyperbolic relationship between the microbial growth rate and the carbon source (organic compound) supporting the microbial growth. Curve represents no inhibitions. Notes: μ is the specific growth rate coefficient; λ is the maximum growth rate coefficient occurring at 0.5 $\mu_{\max}$; K_s is the Monod coefficient; and subscript S is the concentration of limiting nutrient, which can be represented by oxygen demand or total organic carbon [TOC]. *(Data from D.L. Russell, Practical Wastewater Treatment, John Wiley & Sons, 2019.)*

3. Exponential phase.
4. Declining growth phase.
5. Stationary phase.
6. Decay.
7. Exponential death phase.

The cycle returns when the microbes again contact the organic compounds in the food.

Sulfate compounds are reduced to H_2S by microbially mediated reactions:

$$6SO_{2^-} + 37.5H^+ + 30e^- \rightarrow 3.75H_2S + 15H_2O \tag{9.18}$$

In such redox reactions, electrons are transferred from an organic compound in the microbe's food or from the environment to an electron acceptor. In nature, the sediment at the bottom of a waterbody is deep enough to have a layer of very low O_2 concentrations. In an engineered system, it could involve designed O_2 removal. In either case, for an aerobic reaction, O_2 is the usual electron acceptor. Aerobic conditions mean that sufficient molecular oxygen is present. It does not mean, however, that O_2 is the exclusive electron acceptor in all aerobic microbial processes. For example, trivalent iron (Fe^{3+}) as iron sulfate, and carbon dioxide can also be the electron acceptors (see Table 9.1).

The first order biodegradation rate (R) can be calculated empirically from two representative points within the profile of the unconsolidated media:

$$R = \ln \frac{C_d}{C_u}(Dv) \tag{9.19}$$

TABLE 9.1 Redox reactions for benzene, xylene, and ethylbenzene.

Type	Reaction	Electron acceptor
Benzene redox reactions		
Oxidation	$C_6H_6+12H_2O \rightarrow 6CO_2+30H^++30e^-$	Oxygen
Reduction	$7.5O_2+30H^++30e^- \rightarrow 15H_2O$	
Reduction	$6NO_3^-+36H^++30e^- \rightarrow 3N_2+18H_2O$	Nitrate
Reduction	$15Mn^{4+}+30e^- \rightarrow 15Mn^{2+}$	Manganese
Reduction	$30Fe^{3+}+30e^- \rightarrow 30Fe^{2+}$	Iron
Reduction	$6SO_2^-+37.5H^++30e^- \rightarrow 3.75H_2S+15H_2O$	Sulfate
Reduction	$3.75CO_2+30H^++30e^- \rightarrow 3.75CH_4+7.5H_2O$	Methanogenic bacteria
Overall	$C_6H_6+7.5O_2 \rightarrow 6CO_2+3H_2O$	Oxygen
Overall	$C_6H_6+6H^++6NO_3^- \rightarrow 6CO_2+3N_2+6H_2O$	Nitrate
Overall	$C_6H_6+15Mn^{4+}+12H_2O^- \rightarrow 6CO_2+30H^++15Mn^{2+}$	Manganese
Overall	$C_6H_6+30Fe^{3+}+12H_2O^- \rightarrow 6CO_2+30H^++0Fe^{2+}$	Iron
Overall	$C_6H_6+3.75SO_4^{2-}+7.5H^+ \rightarrow 6CO_2+3.75H_2S+3H_2O$	Sulfate
Overall	$C_6H_6+4.5H_2O \rightarrow 2.25CO_2+3.75CH_4$	Methanogenic bacteria
Xylene and ethylbenzene redox reactions		
Oxidation	$C_8H_{10}+16H_2O \rightarrow 8CO_2+42H^++42e^-$	Oxygen
Reduction	$10.5O_2+42H^++42e^- \rightarrow 21H_2O$	
Reduction	$8.4NO_3^-+50.4H^++42e^- \rightarrow 4.2N_2+25.2H_2O$	Nitrate
Reduction	$21Mn^{4+}+42e^- \rightarrow 21Mn^{2+}$	Manganese
Reduction	$42Fe^{3+}+42e^- \rightarrow 42Fe^{2+}$	Iron
Reduction	$5.25SO_2^-+52.5H^++42e^- \rightarrow 5.25H_2S+21H_2O$	Sulfate

TABLE 9.1 Redox reactions for benzene, xylene, and ethylbenzene—cont'd

Type	Reaction	Electron acceptor
Reduction	$5.25CO_2+42H^++42e^- \rightarrow 5.25CH_4+10.5H_2O$	Methanogenic bacteria
Overall	$C_8H_{10}+10.5O_2 \rightarrow 8CO_2+5H_2O$	Oxygen
Overall	$C_8H_{10}+8.4H^++8.4NO_3^- \rightarrow 8CO_2+4.2N_2+9.2H_2O$	Nitrate
Overall	$C_8H_{10}+21Mn^{4+}+16H_2O^- \rightarrow 8CO_2+42H^++21Mn^{2+}$	Manganese
Overall	$C_8H_{10}+42Fe^{3+}+16H_2O^- \rightarrow 8CO_2+42H^++42Fe^{2+}$	Iron
Overall	$C_8H_{10}+5.25SO_4^{2-}+10.5H^+ \rightarrow 8CO_2+5.25H_2S+5H_2O$	Sulfate
Overall	$C_8H_{10}+5.5H_2O \rightarrow 2.75CO_2+5.25CH_4$	Methanogenic bacteria

The three isomers of xylene and ethylbenzene have the same formula, i.e., C_8H_{10}, but the xylenes' methyl group (—CH_3) is located on at three different carbon sites on the benzene ring and ethylbenzene's methyl group is located on a carbon bonded with the ring (—C—CH_3). Thus, the reactions are shown as the same, but the reaction rates differ.
Data from H. Rifai, C. Newell, J. Gonzales, S. Dendrou, L. Kennedy, J. Wilson, BIOPLUME III Natural Attenuation Decision Support System, Version 1.0, User's Manual, Air Force Center for Environmental Excellence (AFCEE), San Antonio, 1997.

Where C_d is the highest downgradient concentration of compound, C_u is the highest upgradient concentration of compound, D is the distance traveled, v is the velocity of the compound mass in the soil, sediment, or other unconsolidated media column.

Prototypical zero-order, first-order, and second-order microbial decay curves are similar Fig. 4.8 in Chapter 4. The initial and final pollution concentrations are equal, but the rates or kinetics of the three systems differ. The rate orders have been derived from experimental data, i.e., idealized curves that must be adapted to the heterogeneous and highly variable conditions of the field [81,88]. Zero-order decay is linear and first and second-order days are curved. However, estimates of expected microbial growth and metabolism and predictions of pollutant degradation rates under real-world conditions will differ from these theoretical decays and those in handbooks based solely on Monod equation, but provide strong clues about the reaction rates [81]. The Monod calculations are often adjusted with laboratory studies that can be scaled up increasingly to represent actual conditions [89]. In addition, computational models that combine fluid dynamics and microbial kinetics are increasingly becoming available to simulate the physics and biochemical reactions with increasing accuracy and representativeness [91,91].

7 Metrics of biotic condition

Healthy ecosystems have diverse and abundant living creatures. Biology and hydrology are key components of an ecosystem's condition. A biodiversity index is calculated as the number of species in area divided by the number of individuals in that area. It can also be described as species richness, which is the number of unique life forms, evenness, which expresses the equitability among the genera, and heterogeneity, which is the dissimilarity among the genera [92].

The diversity and abundance of biological species indicate ecosystem health. Another eco-indicator is the efficiency of the survival and thriving of biota [94,94]. The diversity and abundance of species and ecosystem efficiency must be maintained. Year-over-year generations of populations of organisms must have conditions that support reproduction and growth, i.e., ecosystem sustainability. For example, loss of canopy species in a forest and the impaction of soil may irreversibly affect the root systems of plants and decrease the soil interstices' support of bacterial and fungal species in soil organic matter (SOM). This, in turn, adversely affects the biotic conditions needed by higher-tier trophic levels of organisms.

Biodiversity is defined as the:

> *"... composition, structure, and function (that) determine, and in fact constitute, the biodiversity of an area. Composition has to do with the identity and variety of elements in a collection, and includes species lists and measures of species diversity and genetic diversity. Structure is the physical organization or pattern of a system, from habitat complexity as measured within communities to the pattern of patches and other elements at a landscape scale. Function involves ecological and evolutionary processes, including gene flow, disturbances, and nutrient cycling" [95].*

Expressions of biological community's diversity are relative and merely indicators, i.e., an index, that incorporate many uncertainties to describe the condition of habitat. One common metric of ecological condition is the Shannon-Weiner index of community diversity [96]:

$$H' = \sum_{i=1}^{m} P_i \ln P_i \tag{9.20}$$

This can also be expressed as:

$$H' = -1.44 \sum_{i=1}^{m} \left(\frac{n_i}{N}\right) \ln\left(\frac{n_i}{N}\right) \tag{9.21}$$

Where, H' = index of community diversity; $P_i = n_i/N$; n_i = number (i.e., density) of the ith genera or species; N = total number (i.e., density) of all organisms in the sample; $i = 1, 2\ldots m$; and m = number of genera or species.

The index is based on entropy, incorporating the number of species living in a habitat, i.e., species richness and their relative abundance, i.e., evenness. As the value of H' increases, the ecosystem's overall sustainability or stability also increases. The calculation of community diversity using the Shannon-Weiner index can demonstrate its value in comparing different habitats or changes in the same habitat over time.

Biodiversity example 1

A biology professor has conducted an ecological impact study that measured the abundance of microbes in a stream that receives effluent from an industrial plant. She found a total of 1510 organisms, represented by seven different species. The actual number count of each microbial species in stream community was 10, 50, 75, 125, 200, 350, 750 m L^{-1}. Using Eq. (9.21), the scientist constructed Table 9.2.

Generally, H' values range from about 1.5 to 4.5, but the diversity index in this ecosystem is approximately 2.1. The index is most useful when comparing systems. So, if the stream is 2.1 and surrounding streams are all around 4, there may be a problem. Further investigation is needed if these are biological indicators of the health of the stream and the facility is releasing pollutants that are known to stress ecosystems.

Now consider what happens if the facility installs new pollution control equipment, with every species quadrupling in abundance (highly unlikely, but let us assume it to keep the math simple). The measurements taken a year later are shown in Table 9.3.

TABLE 9.2 Hypothetical species abundance measurements needed to calculate community diversity H', using: $H' = -1.44 \sum_{i=1}^{m} \left(\frac{n_i}{N}\right) \ln\left(\frac{n_i}{N}\right)$.

i	n_i	n_i/N	$-1.44 \ln(n_i/N)$	$-1.44(n_i/N) \ln(n_i/N)$
1	10	0.006623	7.224883	0.047847
2	50	0.033113	4.907292	0.162493
3	75	0.049669	4.323423	0.21474
4	125	0.082781	3.587834	0.297006
5	200	0.13245	2.911028	0.385567
6	350	0.231788	2.105182	0.487956
7	700	0.463576	1.10705	0.513202
Σ	**1510**	**1**		**2.10881**

Data from D.A. Vallero, Biotic physics, in: Methods and Calculations in Environmental Physics, AIP Publishing LLC, Melville, New York, 2022, pp. 9-1–9-40.

TABLE 9.3 Hypothetical species abundance measurements 1 year after new pollution control equipment has been installed.

i	n_i	n_i/N	$-1.44\ \ln(n_i/N)$	$-1.44(n_i/N)\ \ln(n_i/N)$
1	40	0.006623	7.224883	0.047847
2	200	0.033113	4.907292	0.162493
3	300	0.049669	4.323423	0.21474
4	500	0.082781	3.587834	0.297006
5	800	0.13245	2.911028	0.385567
6	1400	0.231788	2.105182	0.487956
7	2800	0.463576	1.10705	0.513202
Σ	**6040**	**1**		**2.10881**

Data from D.A. Vallero, Biotic physics, in: Methods and Calculations in Environmental Physics, AIP Publishing LLC, Melville, New York, 2022, pp. 9-1–9-40.

Biodiversity example 2[a]

Even though the species abundance quadrupled, the index remains 2.1. The index indicates, correctly, that the total species abundance does not affect diversity.

[a]From reference [32].

Biodiversity example 3[b]

Consider now what would happen if the county encouraged more facilities to locate along the stream and several new sources discharged waste into the air and water nearby. A follow-up study is conducted 5 years later indicating that the species density has changed to 3000, 50, 35, 40, 30, 70, and 15 L^{-1} of the same species of microbes as the early study. Table 9.4 shows how the numbers and diversity changed in 5 years.

After 5 years, the actual number of microbes is more than doubling, but the diversity has been drastically reduced ($H'=0.6$ vs 2.1). This likely indicates that conditions are differentially favorable to one species, such as when concentration of a toxic chemical or a change in habitat as when a barrier like a road is built that interrupts migration. The chemical exposure or change in habitat would differentially affect the seven species, with one species appearing to have a much higher tolerance for the insult than the other six.

TABLE 9.4 Hypothetical species abundance measurements 5 years after new pollution sources locate near the stream.

i	n_i	n_i/N	$-1.44\ \ln(n_i/N)$	$-1.44(n_i/N)\ \ln(n_i/N)$
1	3000	0.92593	0.11082	0.10261
2	50	0.01543	6.00668	0.0927
3	35	0.0108	6.52029	0.07044
4	40	0.01235	6.32801	0.07812
5	30	0.00926	6.74227	0.06243
6	70	0.0216	5.52216	0.11931
7	15	0.00463	7.7404	0.03584
Σ	**3240**	**1**		**0.56144**

[b]From reference [32].

In many ecosystem condition assessments, a small number of rare species with only a few representative organisms should not affect a community's biodiversity. Thus, a community may appear healthy when in fact a sensitive, threatened species in being inordinately harmed. This calls for a dominance index like the Simpson index [97]. Again, P_i is the proportion (n/N) of individuals of one species counted in the community (n), which is divided by the total number of individuals in the community (N):

$$D = \frac{1}{\sum_{i=1}^{m} P_i^2} \tag{9.22}$$

Biodiversity example 4

Based on the professor's first community measurement, comparative Simpson indices can be developed (see Table 9.5).

Note that both indices stay the same albeit the species abundance being quadrupled, which indicates that the total species abundance does not affect diversity calculations in either index. However, doubling of microbes drastically reduced H' and increased D. The dominance of Species 1 appears to support the hypothesis that at 5 years ecological conditions began to favor Species 1 differentially.

TABLE 9.5 Calculation of Simpson indices for the previous H′ calculations for three microbial ecology scenarios.

i	n_i	n_i/N	$(n_i/N)^2$	H′
1	10	0.006623	0.000044	
2	50	0.033113	0.001096	
3	75	0.049669	0.002467	
4	125	0.082781	0.006853	
5	200	0.13245	0.017543	
6	350	0.231788	0.053726	
7	700	0.463576	0.214903	
Σ	**1510**	**1**	**0.296631**	**2.10881**
1	40	0.006623	0.000044	
2	200	0.033113	0.001096	
3	300	0.049669	0.002467	
4	500	0.082781	0.006853	
5	800	0.13245	0.017543	
6	1400	0.231788	0.053726	
7	2800	0.463576	0.214903	
Σ	**6040**	**1**	**0.296631**	**2.10881**
1	3000	0.92593	0.857346	
2	50	0.01543	0.000238	
3	35	0.0108	0.000117	
4	40	0.01235	0.000153	
5	30	0.00926	0.000086	
6	70	0.0216	0.000467	
7	15	0.00463	0.000021	
Σ	**3240**	**1**	**0.858427**	**0.56144**

Data from D.A. Vallero, Biotic physics, in: Methods and Calculations in Environmental Physics, AIP Publishing LLC, Melville, New York, 2022, pp. 9-1–9-40.

These and other diversity indices apply to microbial communities, macrophytic plants, fauna, and combinations of species in communities. They can be a first step in evaluating food chains and webs of various genera, functional types, and polymorphs, as well as biomass of a single species. They can be complemented with data regarding ecological conditions such species richness, i.e., the number of different species in a community. Often, it is much easier to observe the number of species than the abundance of species, so richness is often shorthand for biodiversity. As the tables above indicate, however, having but a few individuals suggests ecological vulnerability. For example, a few remaining top carnivores in a community is an indication of a threatened or endangered species, which calls for special protections.

Of course, consistency in measurements and modeling is fundamental to comparing ecosystem conditions among ecological communities. For example, there can be considerable intra-annual and seasonal variability in microbial abundance. Most microbes prefer warmer conditions, but some species compete more effectively in cooler waters. If the studies are comparable, further investigation is needed, but these findings are certainly an indication of a problem given that the expected H' value range is 1.5–4.5. Changes in diversity represent a first eco-risk screen and a forewarning of ecological harm. Generally, situations where diversity as measured by the index show a decline, this could indicate than an ecosystem is experiencing increasing stress.

Another ecosystem condition metric is productivity, i.e., how economical a system uses available energy. Productivity is the amount of biomass that is produced from abiotic resources like nutrients and minerals, and the efficiency of transfer of biotic resources from microbial populations to canopy plant species to top predators. Net primary productivity (NPP or P_I) is the difference between two energy rates:

$$P_I = k_p - k_e \tag{9.23}$$

Where k_p = rate of chemical energy storage by primary producers; and k_e = rate at which the producers use energy via respiration.

Sustainability describes the likelihood that the diversity and productivity can continue at present or improved rates in the future. Thus, predictions of ecological sustainability are difficult, relying on statistical and probabilistic methods. Often the extent of sustainability characterizations and predictions are semi-quantitative, extrapolating current trends of conditions to estimate whether things are improving or degrading, and whether the future conditions are sufficiently uncertain that precautions call for additional protective factors of safety to avoid irreversible problems.

Increasingly, evidence-based risk assessments are being augmented or even supplanted by precaution, especially if the decision involves a reasonable likelihood that an adverse effect is severe and irreversible [98]. The precautionary approach also calls for systems thinking and sustainable solutions. Recently developed screening models and tools, including multicriteria decision analysis (MCDA), give consideration of numerous variables and apply data from various sources. Tools like expert elicitation gather insights from members of specific disciplines within the scientific community on emerging environmental problems [99]. Emerging pollutants are often data-poor, lacking reliable historical data on the characteristics and potential risks of pollutants.

Wetlands provide are instructive of the need for ecosystem condition characterization. Large numbers of species exist and undergo change at the interfaces between surface and ground water, air, soil, sediment, and biota. For example, coastal wetlands like saltwater marshes undergo seasonal and interannual conditions as they are affected by tidal flooding and draining when the high salinity tidal water encroaches on fresher waters (see Fig. 9.6). Wetland habitats support fisheries, providing sustenance, refuge, and nursery habitat for various fisheries species. For coastal marshes in the northeastern regions of the U.S., for example, these include shrimp (order Decapoda), blue crab (*Callinectes sapidu*), and various finfish.

FIG. 9.6 Rhode Island saltwater marsh in Narragansett Bay National Estuarine Research Reserve. *(Photo courtesy of the U.S. National Oceanic and Atmospheric Administration.)*

The soils in wetlands consist of mixes of soil types and moisture. For saltwater marshes these are predominantly mud and peat horizons. Peat forms when decomposed plant biomass accumulates in place. Peat exists in many climate types, including arctic, taiga and boreal peatlands, as well as swamps in temperate and tropical climates [100].

Salt marshes provide protections against dune and other coastline erosion from wave actions, capturing sediment, reducing flooding by acting as a sponge that holds water that would otherwise back up. They play an important role as carbon sinks, continuously absorbing atmospheric carbon and holding it at an average carbon burial rate of 210 g C m^{-2} $year^{-1}$ [101].

Wetland plant biomass continuously undergoes decomposition by microbes. These reactions make for a reduced, low-oxygen environment, i.e., hypoxic, with depth. Tides can submerge marshes for long periods of time, allowing biomass to accumulate as organic matter is decomposed continuously by microbes, particularly bacteria. The decomposition depresses dissolved O_2 concentrations in the interstitial water, so that anaerobic microbial populations increase in the mud and peat that during hypoxic conditions [102]. This induces bacteria-mediated reactions that release CH_4, while more oxidized conditions are releasing CO_2. The anoxic environment also causes oxidized compounds in the plant debris to become reduced. Notably, sulfur in the plant biomass is reduced to H_2S.

In this case, sulfate (SO_4^{2-}) becomes the most abundant electron acceptor. This reaction's release of H_2S is responsible for the rotten-egg odor common to some marshlands. Both oxidation and reduction occur in saltwater marshes, but if normally aerobic or anaerobic microbial processes are interrupted by the introduction of pollutants onto the marsh soil or sediment, the balance can change. The change in redox of wetlands can have climatic impacts. In fact, wetlands hold large amounts carbon so are sinks for CH_4, CO_2, and other gas-phase carbon compounds as their plants and other biota sequester large amounts of CO_2 in underground and sediment biomass. Conversely, loss of wetlands creates a source of CH_4 and CO_2, which is a global greenhouse gas which, when concentrations increase sufficiently, will affect global temperatures and other climatic conditions [103].

Anaerobes in an inundated hypoxic wetland soil decompose carbon compounds more slowly than in higher oxygen soils because aerobic respiration is more thermodynamically efficient than anaerobic respiration. Physical stressors like the disruptions due to dredging and bulldozing of wetlands, causes some of the sequestered carbon to move to the atmosphere [104].

As mentioned, water and air pollutant exposure at sufficiently high concentrations can elicit physiological or biochemical effects in vascular plants., including changes in net photosynthesis rates, stomata response, and metabolic activity. Certain species become unable to survive in an ecosystem even with very subtle changes. For example, plant species exposed to O_3 even at concentrations below current regulatory standards experience decreased growth, and lower biomass production [23,105].

Other sensitive habitats share the biodiversity challenges of wetlands. Tropical rainforest acreage is lost due to clear-cutting and harvesting whole trees [106], as well as the construction of roads. These human activities diminish biodiversity by truncating migration routes and altering feeding zones, as well as increasing soil erosion and decreasing flood control capacity [108,108]. Human activities displace habitats and alter predator-prey, foraging, and breeding behaviors along with other ecosystem functions. About one-fourth of the total greenhouse gas production results from deforestation. Thus, one could argue that the bulldozer can be envisioned as a water and air pollutant, or at least an agent of environmental degradation.

Rainforest species are particularly vulnerable because they adapted to microhabitats that only exist in the unique conditions under ambient moisture and temperature regimes. Other species are lost to habitat destruction and deforestation. Ecosystems edges are particularly important to species survival. When they change due to climatic and anthropogenic stresses, canopy species can no longer survive. This leads to further destruction within the ecosystem, with rapid decreases in species abundance and loss of biodiversity. About half of the earth's species exist in rainforests while they cover merely 7% of the land surface [110,110]. About 25% of the world's total greenhouse gas production comes from deforestation [110].

Soil and plants in tropical forests sequester 460–575 billion metric tons of the earth's surface's carbon. Each square kilometer of tropical forest holds 45,000 tonnes of carbon. From 1850 to 1990, deforestation released 122 billion tonnes of carbon into the atmosphere. Currently the deforestation carbon emission rate is 1.6 billion tonnes per year [111,111].

In addition, deforestation increases emissions of fugitive dust, another example of the hydrosphere-atmosphere-lithosphere-biosphere interactions discussed in Chapter 4. The damage to human health from dust exposure is well documented, so other air-breathing wildlife populations would be expected to be weakened, resulting in decreased species abundance, diminished diversity, and changes to ecosystem function.

Biodiversity also has a social context. For example, individual farmers can suffer at the expense of large-scale corporate farms when the parameters for successful food production only include monetizable values. Indeed, smaller farms with greater biodiversity almost always provide more sustainable agriculture compared to crop monocultures [113,113]. Family farms also have nonmonetizable values, such as livability and psychological benefits to farm families and surrounding communities [114].

Net primary productivity can be a deceiving parameter. Plantation type agriculture consists of one or a few crops year after year, which usually requires large financial investments in fossil fuels to cultivate fields and harvest crops and to apply fossil-fuel based commercial fertilizers, genetically modified seeds, and pesticides. These add up to a lack of diversity and vulnerabilities to infestations and plant diseases [115,116]. Conversely, most small farms are often highly productive with fewer nonsustainable side effects than large-scale operations. For example, corn yields in traditional Mexican *chinampas*, i.e., raised beds in shallow ponds and wetlands, range from 3.5 to 6.3 tons per hectare, which means that up to 20 people can maintain a nutritionally acceptable diet from each hectare farmed [117]. Small farms are also sanctuaries for native species, which help to maintain biodiversity.

8 Conclusions

There are many biological aspects to water quality. This chapter has discussed the biological processes and mechanisms that lead to the formation, transport, transformation, and fate of substances that are or can become water pollutants. This and previous chapters set the stage for the discussions of surface and ground water in the next two chapters.

References

[1] D. Vallero, Environmental Biotechnology: A Biosystems Approach, Elsevier Science, 2015.
[2] D.P. German, M.N. Weintraub, A.S. Grandy, C.L. Lauber, Z.L. Rinkes, S.D. Allison, Optimization of hydrolytic and oxidative enzyme methods for ecosystem studies, Soil Biol. Biochem. 43 (7) (2011) 1387–1397.
[3] J.H. Makoi, P.A. Ndakidemi, Selected soil enzymes: examples of their potential roles in the ecosystem, Afr. J. Biotechnol. 7 (3) (2008).
[4] S.R. Tirkey, S. Ram, S. Mishra, Bacteria-mediated remediation: restoring polycyclic aromatic hydrocarbon (PAHs) contaminated marine ecosystems, in: Advanced Oxidation Processes for Effluent Treatment, Elsevier, Plants, 2021, pp. 119–136.
[5] J. Li, et al., Unraveling bacteria-mediated degradation of lignin-derived aromatic compounds in a freshwater environment, Sci. Total Environ. 749 (2020) 141236.
[6] N.-A.T. Tran, B. Tamburic, C.R. Evenhuis, J.R. Seymour, Bacteria-mediated aggregation of the marine phytoplankton Thalassiosira weissflogii and Nannochloropsis oceanica, J. Appl. Phycol. 32 (6) (2020) 3735–3748.
[7] Y. Li, et al., Coupled carbon, sulfur, and nitrogen cycles mediated by microorganisms in the water column of a shallow-water hydrothermal ecosystem, Front. Microbiol. 9 (2018) 2718.
[8] A. Hüsken, A. Dietz-Pfeilstetter, Pollen-mediated intraspecific gene flow from herbicide resistant oilseed rape (Brassica napus L.), Transgenic Res. 16 (5) (2007) 557–569.
[9] M.A. Rieger, M. Lamond, C. Preston, S.B. Powles, R.T. Roush, Pollen-mediated movement of herbicide resistance between commercial canola fields, Science 296 (5577) (2002) 2386–2388.
[10] R. McKinney, Microbial relationships in biological wastewater treatment systems, in: The structure and function of fresh-water microbial communities, Virginia Polytechnic Institute and State Union, Blacksburg, Virginia, 1973, pp. 165–179.
[11] R.E. McKinney, Discussion of Population Dynamics in Biological Treatment Plants, 1966.
[12] R.E. Mckinney, Microbiology for Sanitary Engineers, McGraw Hill Book Co, New York, 1962.
[13] R.E. McKinney, Civil engineering 876: wastewater treatment plant design, in: Course Lecture, Spring Semester, 1984, University of Kansas, 1984.
[14] D.S. Esser, J.H.J. Leveau, K.M. Meyer, Modeling microbial growth and dynamics, Appl. Microbiol. Biotechnol. 99 (21) (2015) 8831–8846, https://doi.org/10.1007/s00253-015-6877-6.
[15] A. Ahmad, F. Banat, H. Alsafar, S.W. Hasan, Algae biotechnology for industrial wastewater treatment, bioenergy production, and high-value bioproducts, Sci. Total Environ. 806 (2022) 150585.
[16] N.P. Cheremisinoff, Biotechnology for Waste and Wastewater Treatment, Elsevier, 1997.
[17] A. Dragt, J. van Ham, Biotechniques for Air Pollution Abatement and Odour Control Policies, Elsevier, 1992.
[18] P.K. Jain, V.K. Gupta, V. Bajpai, Recent Advances in Environmental Biotechnology, 2011.
[19] National Academies of Sciences, Engineering, and Medicine, Preparing for Future Products of Biotechnology, National Academies Press, 2017.
[20] J. Nielsen, et al., Engineering synergy in biotechnology, Nat. Chem. Biol. 10 (5) (2014) 319–322.
[21] B.E. Rittmann, P.L. McCarty, Environmental Biotechnology: Principles and Applications, Tata McGraw-Hill Education, 2012.
[22] H. Yi, et al., Recent development of advanced biotechnology for wastewater treatment, Crit. Rev. Biotechnol. 40 (1) (2020) 99–118.
[23] D.A. Vallero, Fundamentals of Air Pollution, fifth ed., Elsevier Academic Press, Waltham, MA, 2014. p. 999 pages cm.
[24] S. Mandal, S. Ray, S. Roy, S.E. Jørgensen, Order to chaos and vice versa in an aquatic ecosystem, Ecol. Model. 197 (3) (2006) 498–504, https://doi.org/10.1016/j.ecolmodel.2006.03.020.
[25] R.M. May, Chaos and the dynamics of biological populations, Proc. R. Soc. Lond. A Math. Phys. Sci. 413 (1844) (1987) 27–44.
[26] A. Hastings, T. Powell, Chaos in a three-species food chain, Ecology 72 (3) (1991) 896–903.

[27] T.E. Graedel, On the concept of industrial ecology, Annu. Rev. Energy Environ. 21 (1) (1996) 69–98.
[28] V. Mishra, et al., Health inequalities during COVID-19 and their effects on morbidity and mortality, J. Healthc. Leadersh. (2021) 19–26.
[29] J.B. Awotunde, E.A. Adeniyi, P.O. Kolawole, R.O. Ogundokun, Application of big data in COVID-19 epidemic, in: U. Kose, D. Gupta, V.H.C. de Albuquerque, A. Khanna (Eds.), Data Science for COVID-19, Academic Press, 2022, pp. 141–165 (Chapter 8).
[30] A. Shelke, S. Shelke, S. Acharya, S. Shukla, Synergistic epidemic or syndemic: an emerging pattern of human diseases, Cureus 15 (11) (2023).
[31] D. Vallero, Air Pollution Calculations: Quantifying Pollutant Formation, Transport, Transformation, Fate and Risks, Elsevier, 2019.
[32] D.A. Vallero, Biotic physics, in: Methods and Calculations in Environmental Physics, AIP Publishing LLC, Melville, New York, 2022, pp. 9-1–9-40.
[33] S. Pagad, P. Genovesi, L. Carnevali, R. Scalera, M. Clout, IUCN SSC Invasive Species Specialist Group: invasive alien species information management supporting practitioners, policy makers and decision takers, Manag. Biol. Invasions 6 (2) (2015) 127.
[34] L.E. LaMP, Lake Erie lakewide management plan (LaMP) 2004, in: Report prepared by the Lake Erie LaMP Work Group under the direction of the Lake Erie LaMP Management Committee, Environment Canada, Ontario Region, and the US Environmental Protection Agency, 2004.
[35] National Research Council, Biological Confinement of Genetically Engineered Organisms, National Academies Press, 2004.
[36] A.R. Kapuscinski, Genetic methods for biological control of non-native fish in the Gila River Basin: development and testing of methods, potential environmental risks, regulatory requirements, multi-stakeholder deliberation and cost estimates: final report to the US Fish and Wildlife Service, Institute for Social, Economic and Ecological Sustainability of the University of Minnesota (2005).
[37] M. Hansen, R. DeFries, J. Townshend, M. Carroll, C. Dimiceli, R. Sohlberg, Global percent tree cover at a spatial resolution of 500 meters: first results of the MODIS vegetation continuous fields algorithm, Earth Interact. 7 (10) (2003) 1–15.
[38] R. Lindsey, R. Simmon, Tropical Deforestation, NASA Earth Observatory (Ed.), NASA, Washington, DC, 2007.
[39] M.A. Bruns, K.M. Scow, DNA fingerprinting as a means to identify sources of soil-derived dust: problems and potential, in: Integrated Assessment of Ecosystem Health, CRC Press, 2019, pp. 193–206.
[40] H.D. Scott, Soil Physics: Agricultural and Environmental Applications, Iowa State University Press, Ames, Iowa, 2000.
[41] D.A. Vallero, Methods and Calculations in Environmental Physics, AIP Publishing, Melville, New York, 2022.
[42] Y.X. Kim, K. Ranathunge, S. Lee, Y. Lee, D. Lee, J. Sung, Composite transport model and water and solute transport across plant roots: an update, Front. Plant Sci. 9 (2018) 193.
[43] E. Steudle, C.A. Peterson, How does water get through roots? J. Exp. Bot. 49 (322) (1998) 775–788.
[44] E. Steudle, J. Frensch, Osmotic responses of maize roots, Planta 177 (3) (1989) 281–295.
[45] E. Steudle, J. Frensch, Water transport in plants: role of the apoplast, Plant Soil 187 (1) (1996) 67–79.
[46] M.K. Shukla, Soil Physics: An Introduction, CRC Press, 2013.
[47] Laboratory of Ecohydrology (ECHO), n.d., Chapter 4: Water potential in porous media, ECHO, https://echo2.epfl.ch/VICAIRE/mod_3/chapt_4/main.htm (Accessed July 19, 2024).
[48] H. Bruus, Theoretical Microfluidics, Oxford University Press, 2007.
[49] J. Pfitzner, Poiseuille and his law, Anaesthesia 31 (2) (1976) 273–275.
[50] R. Lewis, S. Gordon, Sampling for organic chemicals in air, Environmental Protection Agency, Research Triangle Park, NC, USA, 1994.
[51] U.-U.T. Võ, M.P. Morris, Nonvolatile, semivolatile, or volatile: Redefining volatile for volatile organic compounds, J. Air Waste Manage. Assoc. 64 (6) (2014) 661–669.
[52] C. McGahey, E. Bouwer, Biodegradation of ethylene glycol in simulated subsurface environments, Water Sci. Technol. 26 (1–2) (1992) 41–49.
[53] A. Ardila-Arias, E. Berrfo-Mesa, E. Arriola-Villaseñor, W. Alverez-Gomez, J. Hernandez-Maldanado, T. Zepeda-Partida, L. Ortiz-Frade, R. - Barrera-Zapata, Degradation of ethylene glycol through photo-Fenton heterogeneous system, Rev. Ing. Univ. Medellín 18 (35) (2019) 91–109, https://doi.org/10.22395/rium.v18n35a6.
[54] K. Ramus, F.-D. Kopinke, A. Georgi, Influence of dissolved humic substances on the mass transfer of organic compounds across the air–water interface, Chemosphere 86 (2) (2012) 138–143, https://doi.org/10.1016/j.chemosphere.2011.09.055.
[55] R.W. Williams, R.R. Watts, R.K. Stevens, C.L. Stone, J. Lewtas, Evaluation of a personal air sampler for twenty-four hour collection of fine particles and semivolatile organics, J. Expo. Sci. Environ. Epidemiol. 9 (2) (1999) 158.
[56] N. Ágnes, K. Rajmund, The environmental impact of plastic waste incineration, Acad. Appl. Res. Mil. Public Manag. Sci. 15 (3) (2016) 231–237.
[57] J. Thornton, M. McCally, P. Orris, J. Weinberg, Hospitals and plastics. Dioxin prevention and medical waste incinerators, Public Health Rep. 111 (4) (1996) 298–313 (in English).
[58] California Office of Environmental Health Hazard Assessment, 2,3,7,8-Tetrachlorodibenzo-p-dioxin and related compounds, 2024. https://oehha.ca.gov/chemicals/2378-tetrachlorodibenzo-p-dioxin-and-related-compounds. (Accessed 19 July 2024).
[59] D.A. Vallero, Air pollution physics, in: Methods and Calculations in Environmental Physics, AIP Publishing LLC, Melville, New York, 2022, pp. 7-1–7-70.
[60] E.J. Kleynhans, A.L. Atchley, S.T. Michaletz, Modeling fire effects on plants: from organs to ecosystems, in: E.A. Johnson, K. Miyanishi (Eds.), Plant Disturbance Ecology, second ed., Academic Press, San Diego, 2021, pp. 383–421 (Chapter 11).
[61] E. Steudle, The cohesion-tension mechanism and the acquisition of water by plant roots, Annu. Rev. Plant Biol. 52 (1) (2001) 847–875.
[62] T. Lawson, M.R. Blatt, Stomatal size, speed, and responsiveness impact on photosynthesis and water use efficiency, Plant Physiol. 164 (4) (2014) 1556–1570.
[63] E.S. Bernhardt, W. Schlesinger, Biogeochemistry: An Analysis of Global Change, Elsevier Inc, Waltham, MA, USA and Oxford, UK, 2013.
[64] S. Wilkinson, G. Mills, R. Illidge, W.J. Davies, How is ozone pollution reducing our food supply? J. Exp. Bot. 63 (2) (2012) 527–536.
[65] G.E. Brurst, Air pollution effects on vegetables, 2018. https://extension.umd.edu/learn/air-pollution-effects-vegetables. (Accessed 25 June 2018).

[66] W.R. Gardner, Dynamic aspects of water availability to plants, Soil Sci. 89 (2) (1960) 63–73.
[67] J.R. Philip, Plant water relations: some physical aspects, Annu. Rev. Plant Physiol. 17 (1) (1966) 245–268.
[68] A.J. McElrone, B. Choat, G.A. Gambetta, C.R. Brodersen, Water uptake and transport in vascular plants, Nat. Educ. Knowl. 4 (6) (2013).
[69] W.H. Schlesinger, E.S. Bernhardt, Biogeochemistry: An Analysis of Global Change, fourth ed., Academic Press, Amsterdam, NV, 2020, ISBN: 978-0128146088.
[70] C.B. Osmond, K. Winter, H. Ziegler, Functional significance of different pathways of CO2 fixation in photosynthesis, in: Physiological Plant Ecology II, Springer, 1982, pp. 479–547.
[71] L. Jeanguenin, A.P. Mir, F. Chaumont, Uptake, loss and control, in: L. Taiz, I.M. Møller, A. Murphy, E. Zeiger (Eds.), Plant Physiology and Development, Elsevier Academic Press, Amsterdam, NV, 2016, pp. 135–140, https://doi.org/10.1016/B978-0-12-394807-6.00087-3. In this issue.
[72] C. Yuan, C. Tebes-Stevens, E.J. Weber, A reaction library to predict direct photochemical transformation products of environmental organic contaminants in sunlit aquatic systems, Environ. Sci. Technol. (2020).
[73] D.A. Vallero, Environmental Systems Science: Theory and Practical Applications, first ed., Elsevier, Amsterdam, NV, 2020.
[74] D.A. Vallero, Environmental Contaminants: Assessment and Control, Elsevier Academic Press, Amsterdam; Boston, 2004. p. xxxix, 801 pp.
[75] D.A. Vallero, C.K. Gunsch, Applications and implications of emerging biotechnologies in environmental engineering, J. Environ. Eng. 146 (6) (2020) 03120005.
[76] M.M. Epstein, T. Vermeire, Scientific opinion on risk assessment of synthetic biology, Trends Biotechnol. 34 (8) (2016) 601–603.
[77] E.U. von Weizsäcker, The environmental dimension of biotechnology, in: Ernst Ulrich von Weizsäcker, Springer, 2014, pp. 57–65.
[78] B.E. Logan, J.R. Hunt, Bioflocculation as a microbial response to substrate limitations, Biotechnol. Bioeng. 31 (2) (1988) 91–101.
[79] J.P. Boltz, E.J. La Motta, J.A. Madrigal, The role of bioflocculation on suspended solids and particulate COD removal in the trickling filter process, J. Environ. Eng. 132 (5) (2006) 506–513.
[80] G. Vaughan, G. Holder, Substrate removal in the trickling filter process, J. Water Pollut. Control Fed. (1984) 417–423.
[81] W. Gujer, M. Boller, Design of a nitrifying tertiary trickling filter based on theoretical concepts, Water Res. 20 (11) (1986) 1353–1362, https://doi.org/10.1016/0043-1354(86)90133-8.
[82] M.J. Higgins, J.T. Novak, Characterization of exocellular protein and its role in bioflocculation, J. Environ. Eng. 123 (5) (1997) 479–485.
[83] X. Zhang, P.L. Bishop, B.K. Kinkle, Comparison of extraction methods for quantifying extracellular polymers in biofilms, Water Sci. Technol. 39 (7) (1999) 211–218.
[84] W.P. Gerhard, Sanitation and Sanitary Engineering, The Author, 1909.
[85] W. Eckenfelder, Biological treatment of waste water, Adv. Hydrosci. 3 (1966) 153–190.
[86] D.A. Vallero, J.J. Peirce, Engineering the Risks of Hazardous Wastes, Butterworth-Heinemann, Burlington, MA, 2003. p. xxi, 306 pp.
[87] M.P. Lau, M. Sander, J. Gelbrecht, M. Hupfer, Natural organic matter as electron acceptor: experimental evidence for its important role in anaerobic respiration, in: EGU General Assembly Conference Abstracts, 2014, p. 6136.
[88] S. Liu, How cells grow, in: S. Liu (Ed.), Bioprocess Engineering, Elsevier, 2017, pp. 629–697. second ed. (Chapter 11).
[89] K. van't Riet, R.G.J.M. van der Lans, 2.07—Mixing in bioreactor vessels, in: M. Moo-Young (Ed.), Comprehensive Biotechnology, second ed., Academic Press, Burlington, 2011, pp. 63–80.
[90] X. Zou, J.-y. Xia, J. Chu, Y.-p. Zhuang, S.-l. Zhang, Real-time fluid dynamics investigation and physiological response for erythromycin fermentation scale-up from 50 L to 132 m3 fermenter, Bioprocess Biosyst. Eng. 35 (5) (2012) 789–800.
[91] J. Bao, Q. Ye, J.-J. Zhong, Bioreactor Engineering Research and Industrial Applications II, Springer, 2016.
[92] Q. Yang, G. Liu, M. Casazza, F. Gonella, Z. Yang, Three dimensions of biodiversity: new perspectives and methods, Ecol. Indic. 130 (2021) 108099, https://doi.org/10.1016/j.ecolind.2021.108099.
[93] S.T. Allen, et al., Ecological resilience indicators for salt marsh ecosystems, in: Ecological Resilience Indicators for Five Northern Gulf of Mexico Ecosystems, NatureServe, Arlington, 2018, p. 53.
[94] B. Hermann, C. Kroeze, W. Jawjit, Assessing environmental performance by combining life cycle assessment, multi-criteria analysis and environmental performance indicators, J. Clean. Prod. 15 (18) (2007) 1787–1796.
[95] R.F. Noss, Indicators for monitoring biodiversity: a hierarchical approach, Conserv. Biol. 4 (4) (1990) 355–364.
[96] W. Strong, Biased richness and evenness relationships within Shannon–Wiener index values, Ecol. Indic. 67 (2016) 703–713.
[97] P.J. Somerfield, K.R. Clarke, R.M. Warwick, Simpson index, in: S.E. Jørgensen, B.D. Fath (Eds.), Encyclopedia of Ecology, Academic Press, Oxford, 2008, pp. 3252–3255.
[98] United Nations Environment Programme, Rio declaration on environment and development, Nairobi, Kenya, 1992.
[99] M.D. Wood, K. Plourde, S. Larkin, P.P. Egeghy, A.J. Williams, V. Zemba, I. Linkov, D.A. Vallero, Advances on a decision analytic approach to exposure-based chemical prioritization, Risk Anal. 40 (1) (2020) 83–96.
[100] Canadian Agricultural Services Coordinating Committee, Soil Classification Working Group, National Research Council Canada, Canada. Agriculture, Agri-Food Canada. Research Branch, The Canadian system of soil classification (no. 1646), NRC Research Press, Ottawa, Canada, 1998.
[101] J. Bai, et al., Depth-distribution patterns and control of soil organic carbon in coastal salt marshes with different plant covers, Sci. Rep. 6 (1) (2016) 1–12.
[102] R. Conrad, Methane production in soil environments—anaerobic biogeochemistry and microbial life between flooding and desiccation, Microorganisms 8 (6) (2020) 881.
[103] R.R. Lane, S.K. Mack, J.W. Day, R.D. DeLaune, M.J. Madison, P.R. Precht, Fate of soil organic carbon during wetland loss, Wetlands 36 (2016) 1167–1181.

[104] R.H. Day, S.T. Allen, J. Brenner, K. Goodin, D. Faber-Langendoen, K.W. Ames, Ecological resilience indicators for mangrove ecosystems, in: Ecological Resilience Indicators for Five Northern Gulf of Mexico Ecosystems, NatureServe, Arlington, Virginia, USA, 2018.
[105] C.S.B.W.W. Heck, Effects on vegetation, in: A.C. Stern (Ed.), Air Pollution, third ed., Academic Press, New York, NY, 1977, pp. 157–229.
[106] L. Bennett, Deforestation and Climate Change, vol. 1400, A Publication of Climate Institute, 2017.
[107] G. Ofomata, Impact of road building, urbanisation and general infrastructural development on the Nigerian rainforest ecosystem, Landsc. Plan. 8 (1) (1981) 21–29.
[108] S.C. Trombulak, C.A. Frissell, Review of ecological effects of roads on terrestrial and aquatic communities, Conserv. Biol. 14 (1) (2000) 18–30.
[109] D.A. Vallero, T.M. Letcher, Unraveling Environmental Disasters, Newnes, 2012.
[110] R. Lindsey, Tropical Deforestation. March 30, 2007; Adapted From R. Simmon. (2005). Millennium Ecosystem Assessment Biodiversity Synthesis, Goddard Space Flight Center, Earth Observatory, U.S. National Aeronautics and Space Administration, 2007. https://earthobservatory.nasa.gov/features/Deforestation. (Accessed 27 October 2021).
[111] NASA, Measurements of Pollution in the Troposphere (MOPITT) Data Langley Data Center Collection Guide, 2000, Available: http://eosweb.larc.nasa.gov.
[112] C.A. Monteiro, G. Cannon, The impact of transnational "big food" companies on the south: a view from Brazil, PLoS Med. 9 (7) (2012) e1001252.
[113] R. Patel, How society subsidizes big food and poor health, JAMA Intern. Med. 176 (8) (2016) 1132–1133, https://doi.org/10.1001/jamainternmed.2016.3068.
[114] K.M. Siegal, The tentative promise of social return on investment, Am. J. Eval. 43 (3) (2022) 438–457, https://doi.org/10.1177/10982140211072420. In this issue.
[115] D.A. Vallero, T.M. Letcher, Engineering risks and failures: lessons learned from environmental disasters, Leadersh. Manag. Eng. 12 (4) (2012) 199–209.
[116] D.A. Vallero, T.M. Letcher, Unraveling Environmental Disasters, second ed., Elsevier Academic Press, Amsterdam, The Netherlands, 2023.
[117] M.A. Altieri, Agroecology, small farms, and food sovereignty, Mon. Rev. 61 (3) (2009) 102–113.